

Heeseung Choi

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Research Interests

My research interests include, but are not limited to:

- RF/Microwave Circuits & Front-End Systems
- Applied Electromagnetics
- Wireless Communication

Education

Yonsei University, Seoul, South Korea

03/2020 – 02/2026

Bachelor of Engineering in Electrical and Electronic Engineering

GPA: 4.07/4.30 (including mandatory military service)

University of Georgia, GA, United States

08/2024 – 12/2024

Exchange Program

GPA: 4.00/4.00

Research Experience

Undergraduate Researcher

Seoul, South Korea

Yonsei University — Advisor: Prof. Byung-Wook Min

01/2025 – Present

- Led an undergraduate capstone on a **1-bit Phase Shifter Module** for Reconfigurable Intelligent Surface (RIS), achieving a stable 180° phase shift and low insertion loss between two states over a wide bandwidth.
- Extending the 1-bit design to **1.5-/2-bit Phase Shifter Modules** for improved phase resolution and beam control via additional states; and designing 8-channel dual-polarized RIS layout for fabrication.
- Awarded the IEEE AP-S Undergraduate Research Scholarship; won 1st Place in the Yonsei E.E. Graduation Project Competition; and presented at KIEES (2025).

Awards & Scholarships

- **IEEE AP-S Undergraduate Summer Research Scholarship**

IEEE Antennas and Propagation Society (IEEE AP-S)

Selected among 102 applicants based on a proposal titled “A Wideband Reconfigurable Intelligent Surface using 1-bit GaAs HEMT Phase Shifter”.

- **1st Place, Graduation Project Competition**

Yonsei University, Department of Electrical and Electronic Engineering

Secured 1st place among 67 participating teams based on the evaluation by faculty members.

- **Korea-U.S. Advanced Technology Exchange Scholarship**

Korea Institute for Advancement of Technology (KIAT)

Awarded to 90 top engineering students for U.S. exchange programs, based on excellence and potential.

- **Academic Honors (Dean's List):** 2020, 2023, 2024, 2025

Yonsei University

- **Yonsei University Semiconductor Track Program**

Yonsei University

Publications & Presentations

- [1] **Heeseung Choi** and Byung-Wook Min, “1-bit Phase-Shifting Switch Module for Reconfigurable Intelligent Surfaces,” *KIEES Summer Conference*, Pyeongchang, South Korea, Aug. 2025 (Poster Presentation).

Skills

- **Tools:** Keysight ADS, Ansys HFSS, Cadence AWR, MATLAB, PSPICE/LTSPICE, Vivado
- **RF Measurement:** VNA, SA, VSG, Anechoic Chamber
- **Relevant Courses:** Electronic Circuits I/II, Electricity & Magnetism I/II, Microwave Engineering, Antenna Engineering, Communication Theory, Communication Systems, Digital Signal Processing